INCH-POUND

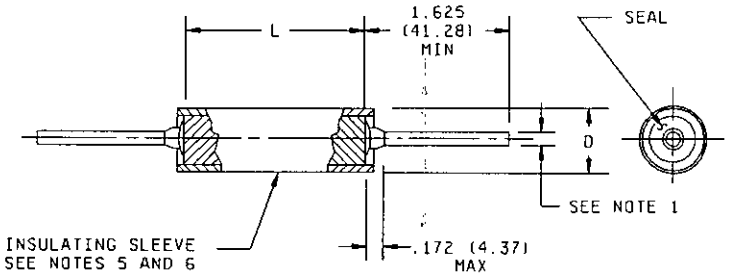
MIL-C-39022/20
28 July 1993
SUPERSEDING
MIL-C-39022/2C
12 October 1970

MILITARY SPECIFICATION SHEET

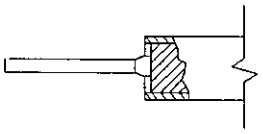
CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM, DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CHR19 (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

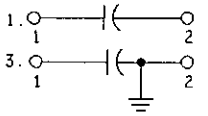
The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39022.



CASE FOR CIRCUIT 1



CASE FOR CIRCUIT 3



CIRCUIT DIAGRAM

(D) NOTES:
1. Number 22 AWG wire .025 ±.002 inch (0.64 ±0.05 mm) for case diameters of .235 inch (5.97 mm) and .312 inch (7.92 mm).
   Number 20 AWG wire .032 ±.002 inch (0.81 ±0.05 mm) for case diameters of .400 inch (10.16 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 inch (0.79 mm) on either end. Insulating sleeve thickness shall not exceed .016 inch (0.41 mm).
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.
8. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Dimensions and configuration.

(D) denotes change

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: ±5 percent and ±10 percent.

Rated voltage: See table I.

Dielectric material: Normally paper-polyethylene-terephthalate.

Operating temperature range: -55°C to +125°C.

(D) Failure rate level: M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent).

Radiographic inspection (X-ray): Method 209 of MIL-STD-202.

(D) Thermal shock: Method 107 of MIL-STD-202, condition A, except step 3 shall be +125°C +4°C, -0°C.

Seal: Method 112 of MIL-STD-202, condition A, or (for liquid-filled units only) 1 hour at +125°C +4°C, -0°C with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.

Terminal to terminal: 200 percent of rated voltage.

Terminals to case (when case is not a terminal): 200 percent of rated voltage.

NOTE: For 100 percent inspection:

250 percent for not less than 5 seconds, or

200 percent for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.

Terminal to terminal: See table II.

Terminals to case (when case is not a terminal): 10,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF): 1 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).

125 percent of rated voltage applied: See MIL-C-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's peak).

50 percent of rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: 5 percent.

Moisture resistance: Method 106 of MIL-STD-202.

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(D) TABLE I. Electrical characteristics, dimensions and dash numbers.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1,000 hr)			
Volts	µF		Percent ±	L ±0.062	D ±0.031	M(1.0)	P(0.1)	R(0.01)	S(0.001)
200	0.10	1	5	1.125	0.312	1111	1221	1331	1441
200	0.10	1	10	1.125	0.312	1112	1222	1332	1442
200	0.10	3	5	1.062	0.312	1113	1223	1333	1443
200	0.10	3	10	1.062	0.312	1114	1224	1334	1444
200	0.15	1	5	1.125	0.312	1115	1225	1335	1445
200	0.15	1	10	1.125	0.312	1116	1226	1336	1446
200	0.15	3	5	1.062	0.312	1117	1227	1337	1447
200	0.15	3	10	1.062	0.312	1118	1228	1338	1448
200	0.22	1	5	1.125	0.400	1119	1229	1339	1449
200	0.22	1	10	1.125	0.400	1120	1230	1340	1450
200	0.22	3	5	1.062	0.400	1121	1231	1341	1451
200	0.22	3	10	1.062	0.400	1122	1232	1342	1452
200	0.33	1	5	1.125	0.500	1123	1233	1343	1453
200	0.33	1	10	1.125	0.500	1124	1234	1344	1454
200	0.33	3	5	1.062	0.500	1125	1235	1345	1455
200	0.33	3	10	1.062	0.500	1126	1236	1346	1456
200	0.47	1	5	1.125	0.500	1127	1237	1347	1457
200	0.47	1	10	1.125	0.500	1128	1238	1348	1458
200	0.47	3	5	1.062	0.500	1129	1239	1349	1459
200	0.47	3	10	1.062	0.500	1130	1240	1350	1460
200	0.68	1	5	1.375	0.562	1131	1241	1351	1461
200	0.68	1	10	1.375	0.562	1132	1242	1352	1462
200	0.68	3	5	1.312	0.562	1133	1243	1353	1463
200	0.68	3	10	1.312	0.562	1134	1244	1354	1464
200	1.0	1	5	1.625	0.562	1135	1245	1355	1465
200	1.0	1	10	1.625	0.562	1136	1246	1356	1466
200	1.0	3	5	1.562	0.562	1137	1247	1357	1467
200	1.0	3	10	1.562	0.562	1138	1248	1358	1468
200	1.5	1	5	1.844	0.562	1139	1249	1359	1469
200	1.5	1	10	1.844	0.562	1140	1250	1360	1470
200	1.5	3	5	1.781	0.562	1141	1251	1361	1471
200	1.5	3	10	1.781	0.562	1142	1252	1362	1472
200	2.0	1	5	1.844	0.672	1143	1253	1363	1473
200	2.0	1	10	1.844	0.672	1144	1254	1364	1474
200	2.0	3	5	1.781	0.672	1145	1255	1365	1475
200	2.0	3	10	1.781	0.672	1146	1256	1366	1476
200	3.0	1	5	2.125	0.750	1147	1257	1367	1477
200	3.0	1	10	2.125	0.750	1148	1258	1368	1478
200	3.0	3	5	2.062	0.750	1149	1259	1369	1479
200	3.0	3	10	2.062	0.750	1150	1260	1370	1480
200	4.3	1	5	2.625	0.750	1151	1261	1371	1486
200	4.3	1	10	2.625	0.750	1152	1262	1372	1482
200	4.3	3	5	2.562	0.750	1153	1263	1373	1483
200	4.3	3	10	2.562	0.750	1154	1264	1374	1484
200	5.1	1	5	1.875	1.000	1155	1265	1375	1485
200	5.1	1	10	1.875	1.000	1156	1266	1376	1486
200	5.1	3	5	1.812	1.000	1157	1267	1377	1487
200	5.1	3	10	1.812	1.000	1158	1268	1378	1488
200	6.2	1	5	2.125	1.000	1159	1269	1379	1489
200	6.2	1	10	2.125	1.000	1160	1270	1380	1490
200	6.2	3	5	2.062	1.000	1161	1271	1381	1491
200	6.2	3	10	2.062	1.000	1162	1272	1382	1492

See footnotes at end of table.

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(D) TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1,000 hr)			
				L	D				
Volts	μF		Percent ±	±0.062	±0.031	M(1.0)	P(0.1)	R(0.01)	S(0.001)
200	8.2	1	5	2.625	1.000	1163	1273	1383	1493
200	8.2	1	10	2.625	1.000	1164	1274	1384	1494
200	8.2	3	5	2.562	1.000	1165	1275	1385	1495
200	8.2	3	10	2.562	1.000	1166	1276	1386	1496
400	0.047	1	5	1.125	0.400	1167	1277	1387	1497
400	0.047	1	10	1.125	0.400	1168	1278	1388	1498
400	0.047	3	5	1.062	0.400	1169	1279	1389	1499
400	0.047	3	10	1.062	0.400	1170	1280	1390	1500
400	0.068	1	5	1.125	0.400	1171	1281	1391	1501
400	0.068	1	10	1.125	0.400	1172	1282	1392	1502
400	0.068	3	5	1.062	0.400	1173	1283	1393	1503
400	0.068	3	10	1.062	0.400	1174	1284	1394	1504
400	0.1	1	5	1.125	0.500	1175	1285	1395	1505
400	0.1	1	10	1.125	0.500	1176	1286	1396	1506
400	0.1	3	5	1.062	0.500	1177	1287	1397	1507
400	0.1	3	10	1.062	0.500	1178	1288	1398	1508
400	0.15	1	5	1.125	0.562	1179	1289	1399	1509
400	0.15	1	10	1.125	0.562	1180	1290	1400	1510
400	0.15	3	5	1.062	0.562	1181	1291	1401	1511
400	0.15	3	10	1.062	0.562	1182	1292	1402	1512
400	0.22	1	5	1.375	0.562	1183	1293	1403	1513
400	0.22	1	10	1.375	0.562	1184	1294	1404	1514
400	0.22	3	5	1.312	0.562	1185	1295	1405	1515
400	0.22	3	10	1.312	0.562	1186	1296	1406	1516
400	0.33	1	5	1.375	0.672	1187	1297	1407	1517
400	0.33	1	10	1.375	0.672	1188	1298	1408	1518
400	0.33	3	5	1.312	0.672	1189	1299	1409	1519
400	0.33	3	10	1.312	0.672	1190	1300	1410	1520
400	0.47	1	5	1.625	0.672	1191	1301	1411	1521
400	0.47	1	10	1.625	0.672	1192	1302	1412	1522
400	0.47	3	5	1.562	0.672	1193	1303	1413	1523
400	0.47	3	10	1.562	0.672	1194	1304	1414	1524
400	0.68	1	5	1.625	0.750	1195	1305	1415	1525
400	0.68	1	10	1.625	0.750	1196	1306	1416	1526
400	0.68	3	5	1.562	0.750	1197	1307	1417	1527
400	0.68	3	10	1.562	0.750	1198	1308	1418	1528
400	1.0	1	5	1.875	0.750	1199	1309	1419	1529
400	1.0	1	10	1.875	0.750	1200	1310	1420	1530
400	1.0	3	5	1.812	0.750	1201	1311	1421	1531
400	1.0	3	10	1.812	0.750	1202	1312	1422	1532
400	1.5	1	5	1.625	1.000	1203	1313	1423	1533
400	1.5	1	10	1.625	1.000	1204	1314	1424	1534
400	1.5	3	5	1.562	1.000	1205	1315	1425	1535
400	1.5	3	10	1.562	1.000	1206	1316	1426	1536
400	2.2	1	5	2.125	1.000	1207	1317	1427	1537
400	2.2	1	10	2.125	1.000	1208	1318	1428	1538
400	2.2	3	5	2.062	1.000	1209	1319	1429	1539
400	2.2	3	10	2.062	1.000	1210	1320	1430	1540
400	3.3	1	5	2.625	1.000	1211	1321	1431	1541
400	3.3	1	10	2.625	1.000	1212	1322	1432	1542
400	3.3	3	5	2.562	1.000	1213	1323	1433	1543
400	3.3	3	10	2.562	1.000	1214	1324	1434	1544

1/ Dimensions are bare case sizes (see figure 1).
2/ Metric equivalents, see table III.

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TABLE II. Insulation resistance.

	At dc voltage of	
	200 V	400 V
In megohms:		
At +25°C ±3°C (need not exceed)	12,000	12,000
At +125°C +4°C, -0°C (need not exceed)	150	600
In megohms x microfarads (minimum):		
At +25°C ±3°C	2,000	2,000
At +125°C +4°C, -0°C	10	40

TABLE III. Metric equivalents of decimal inches. 1/

Inches	mm	Inches	mm	Inches	mm
.031	0.79	1.000	25.40	1.812	46.02
.062	1.57	1.062	26.97	1.844	46.84
.312	7.92	1.125	28.58	1.875	47.63
.400	10.16	1.312	33.32	2.062	52.37
.500	12.70	1.375	34.93	2.125	53.98
.562	14.27	1.562	39.67	2.562	65.07
.672	17.07	1.625	41.28	2.625	66.68
.750	19.05	1.781	45.24		

1/ Metric equivalents are given for general information only.

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Insulating sleeves: Greater than 4,000 volts, dc.

Terminal to terminal: 200 percent of rated voltage.

Terminals to case (when case is not a terminal): 200 percent of rated voltage.

IR:

Insulating sleeves: 100 megohms, minimum.

Terminal to terminal: Not less than 50 percent of value specified in table II.

Terminals to case (when case is not a terminal): 5,000 megohms, minimum.

Cap.: Within ±10 percent of initial value.

DF: Not more than 1.1 percent.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I.

50 percent of rated voltage applied.

DWV, IR, Cap., and DF: Same as for immersion.

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Terminal strength: Method 211 of MIL-STD-202, conditions A and D.

Condition A: Applied force 5 pounds.

Condition D: 3 rotations of 360 degrees.

Low temperature and cap. change with temperature:

Low temperature: -55°C +0°C, -3°C for 48 ±4 hours with rated voltage applied.

Cap. change with temperature: At -55°C +0°C, -3°C: -10 percent, maximum. At +125°C +4°C, -0°C: +20 percent maximum.

Fungus: Method 508 of MIL-STD-810, procedure I.

(D) Resistance to soldering heat: Method 210 of MIL-STD-202, condition G (260°C ±5°C for 10 ±1 seconds).

IR: See table II.

Cap.: Within ±5 percent of initial value.

DF: Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.

Qualification:

Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.

Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.

DF (at +125°C +4°C, -0°C) between 24 and 48 hours of test: 2 percent, maximum.

During last 48 hours of test: 2.5 percent, maximum.

IR:

Insulating sleeves: 100 megohms, minimum.

Terminal to terminal: Not less than 33.3 percent of initial requirement.

Terminals to case (when case is not a terminal): 5,000 megohms, minimum.

Cap.: Within ±10 percent of initial value.

DF (at +25°C ±3°C after life): 1.25 percent.

Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:

Accelerated conditions: 2,000 +72, -0 hours.

Rated conditions: 10,000 +96, -0 hours.

Substitutability: See table IV.

Marking: In accordance with MIL-C-39022.

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(D) TABLE IV. Substitutability data.

Dash number in MIL-C-39022/2D				Dash number in MIL-C-39022/2C					Type designation in MIL-C-39022/2B 1/	Type designation in MIL-C-39022/2A and /2 1/ 2/	Dash number in MIL-C-39022/2
M(1.0)	P(0.1)	R(0.01)	S(0.001)	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)			
1111	1221	1331	1441	1001	1111	1221	1331	1441	CHR19A1NC104--	CHR19A1NC104--	0001
1112	1222	1332	1442	1002	1112	1222	1332	1442	CHR19A1NC104--	CHR19A1NC104--	0002
1113	1223	1333	1443	1003	1113	1223	1333	1443	CHR19A3NC104--	CHR19A3NC104--	0003
1114	1224	1334	1444	1004	1114	1224	1334	1444	CHR19A3NC104--	CHR19A3NC104--	0004
1115	1225	1335	1445	1005	1115	1225	1335	1445	CHR19A1NC154--		
1116	1226	1336	1446	1006	1116	1226	1336	1446	CHR19A1NC154--		
1117	1227	1337	1447	1007	1117	1227	1337	1447	CHR19A3NC154--		
1118	1228	1338	1448	1008	1118	1228	1338	1448	CHR19A3NC154--		
1119	1229	1339	1449	1009	1119	1229	1339	1449	CHR19A1NC224--	CHR19A1NC224--	0005
1120	1230	1340	1450	1010	1120	1230	1340	1450	CHR19A1NC224--	CHR19A1NC224--	0006
1121	1231	1341	1451	1011	1121	1231	1341	1451	CHR19A3NC224--	CHR19A3NC224--	0007
1122	1232	1342	1452	1012	1122	1232	1342	1452	CHR19A3NC224--	CHR19A3NC224--	0008
1123	1233	1343	1453	1013	1123	1233	1343	1453	CHR19A1NC334--		
1124	1234	1344	1454	1014	1124	1234	1344	1454	CHR19A1NC334--		
1125	1235	1345	1455	1015	1125	1235	1345	1455	CHR19A3NC334--		
1126	1236	1346	1456	1016	1126	1236	1346	1456	CHR19A3NC334--		
1127	1237	1347	1457	1017	1127	1237	1347	1457	CHR19A1NC474--	CHR19A1NC474--	0009
1128	1238	1348	1458	1018	1128	1238	1348	1458	CHR19A1NC474--	CHR19A1NC474--	0010
1129	1239	1349	1459	1019	1129	1239	1349	1459	CHR19A3NC474--	CHR19A3NC474--	0011
1130	1240	1350	1460	1020	1130	1240	1350	1460	CHR19A3NC474--	CHR19A3NC474--	0012
1131	1241	1351	1461	1021	1131	1241	1351	1461	CHR19A1NC684--		
1132	1242	1352	1462	1022	1132	1242	1352	1462	CHR19A1NC684--		
1133	1243	1353	1463	1023	1133	1243	1353	1463	CHR19A3NC684--		
1134	1244	1354	1464	1024	1134	1244	1354	1464	CHR19A3NC684--		
1135	1245	1355	1465	1025	1135	1245	1355	1465	CHR19A1NC105--	CHR19A1NC105--	0013
1136	1246	1356	1466	1026	1136	1246	1356	1466	CHR19A1NC105--	CHR19A1NC105--	0014
1137	1247	1357	1467	1027	1137	1247	1357	1467	CHR19A3NC105--	CHR19A3NC105--	0015
1138	1248	1358	1468	1028	1138	1248	1358	1468	CHR19A3NC105--	CHR19A3NC105--	0016
1139	1249	1359	1469	1029	1139	1249	1359	1469	CHR19A1NC155--	CHR19A1NC155--	0017
1140	1250	1360	1470	1030	1140	1250	1360	1470	CHR19A1NC155--	CHR19A1NC155--	0018
1141	1251	1361	1471	1031	1141	1251	1361	1471	CHR19A3NC155--	CHR19A3NC155--	0019
1142	1252	1362	1472	1032	1142	1252	1362	1472	CHR19A3NC155--	CHR19A3NC155--	0020
1143	1253	1363	1473	1033	1143	1253	1363	1473	CHR19A1NC205--	CHR19A1NC205--	0021
1144	1254	1364	1474	1034	1144	1254	1364	1474	CHR19A1NC205--	CHR19A1NC205--	0022
1145	1255	1365	1475	1035	1145	1255	1365	1475	CHR19A3NC205--	CHR19A3NC205--	0023
1146	1256	1366	1476	1036	1146	1256	1366	1476	CHR19A3NC205--	CHR19A3NC205--	0024
1147	1257	1367	1477	1037	1147	1257	1367	1477	CHR19A1NC305--	CHR19A1NC305--	0025
1148	1258	1368	1478	1038	1148	1258	1368	1478	CHR19A1NC305--	CHR19A1NC305--	0026
1149	1259	1369	1479	1039	1149	1259	1369	1479	CHR19A3NC305--	CHR19A3NC305--	0027
1150	1260	1370	1480	1040	1150	1260	1370	1480	CHR19A3NC305--	CHR19A3NC305--	0028
1151	1261	1371	1486	1041	1151	1261	1371	1486	CHR19A1NC435--		
1152	1262	1372	1482	1042	1152	1262	1372	1482	CHR19A1NC435--		
1153	1263	1373	1483	1043	1153	1263	1373	1483	CHR19A3NC435--		
1154	1264	1374	1484	1044	1154	1264	1374	1484	CHR19A3NC435--		
1155	1265	1375	1485	1045	1155	1265	1375	1485	CHR19A1NC515--		
1156	1266	1376	1486	1046	1156	1266	1376	1486	CHR19A1NC515--		
1157	1267	1377	1487	1047	1157	1267	1377	1487	CHR19A3NC515--		
1158	1268	1378	1488	1048	1158	1268	1378	1488	CHR19A3NC515--		
1159	1269	1379	1489	1049	1159	1269	1379	1489	CHR19A1NC625--		
1160	1270	1380	1490	1050	1160	1270	1380	1490	CHR19A1NC625--		
1161	1271	1381	1491	1051	1161	1271	1381	1491	CHR19A3NC625--		
1162	1272	1382	1492	1052	1162	1272	1382	1492	CHR19A3NC625--		

See footnotes at end of table.

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(D) TABLE IV. Substitutability data - Continued.

Dash number in MIL-C-39022/2D				Dash number in MIL-C-39022/2C					Type designation in MIL-C-39022/2B 1/	Type designation in MIL-C-39022/2A and /2 1/ 2/	Dash number in MIL-C-39022/2
M(1.0)	P(0.1)	R(0.01)	S(0.001)	L(5.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)			
1163	1273	1383	1493	1053	1163	1273	1383	1493	CHR19A1NC825--		
1164	1274	1384	1494	1054	1164	1274	1384	1494	CHR19A1NC825--		
1165	1275	1385	1495	1055	1165	1275	1385	1495	CHR19A3NC825--		
1166	1276	1386	1496	1056	1166	1276	1386	1496	CHR19A3NC825--		
1167	1277	1387	1497	1057	1167	1277	1387	1497	CHR19A1NE473--	CHR19A1NE473--	0101
1168	1278	1388	1498	1058	1168	1278	1388	1498	CHR19A1NE473--	CHR19A1NE473--	0102
1169	1279	1389	1499	1059	1169	1279	1389	1499	CHR19A3NE473--	CHR19A3NE473--	0103
1170	1280	1390	1500	1060	1170	1280	1390	1500	CHR19A3NE473--	CHR19A3NE473--	0104
1171	1281	1391	1501	1061	1171	1281	1391	1501	CHR19A1NE683--		
1172	1282	1392	1502	1062	1172	1282	1392	1502	CHR19A1NE683--		
1173	1283	1393	1503	1063	1173	1283	1393	1503	CHR19A3NE683--		
1174	1284	1394	1504	1064	1174	1284	1394	1504	CHR19A3NE683--		
1175	1285	1395	1505	1065	1175	1285	1395	1505	CHR19A1NE104--	CHR19A1NE104--	0105
1176	1286	1396	1506	1066	1176	1286	1396	1506	CHR19A1NE104--	CHR19A1NE104--	0106
1177	1287	1397	1507	1067	1177	1287	1397	1507	CHR19A3NE104--	CHR19A3NE104--	0107
1178	1288	1398	1508	1068	1178	1288	1398	1508	CHR19A3NE104--	CHR19A3NE104--	0108
1179	1289	1399	1509	1069	1179	1289	1399	1509	CHR19A1NE154--		
1180	1290	1400	1510	1070	1180	1290	1400	1510	CHR19A1NE154--		
1181	1291	1401	1511	1071	1181	1291	1401	1511	CHR19A3NE154--		
1182	1292	1402	1512	1072	1182	1292	1402	1512	CHR19A3NE154--		
1183	1293	1403	1513	1073	1183	1293	1403	1513	CHR19A1NE224--	CHR19A1NE224--	0109
1184	1294	1404	1514	1074	1184	1294	1404	1514	CHR19A1NE224--	CHR19A1NE224--	0110
1185	1295	1405	1515	1075	1185	1295	1405	1515	CHR19A3NE224--	CHR19A3NE224--	0111
1186	1296	1406	1516	1076	1186	1296	1406	1516	CHR19A3NE224--	CHR19A3NE224--	0112
1187	1297	1407	1517	1077	1187	1297	1407	1517	CHR19A1NE334--		
1188	1298	1408	1518	1078	1188	1298	1408	1518	CHR19A1NE334--		
1189	1299	1409	1519	1079	1189	1299	1409	1519	CHR19A3NE334--		
1190	1300	1410	1520	1080	1190	1300	1410	1520	CHR19A3NE334--		
1191	1301	1411	1521	1081	1191	1301	1411	1521	CHR19A1NE474--	CHR19A1NE474--	0113
1192	1302	1412	1522	1082	1192	1302	1412	1522	CHR19A1NE474--	CHR19A1NE474--	0114
1193	1303	1413	1523	1083	1193	1303	1413	1523	CHR19A3NE474--	CHR19A3NE474--	0115
1194	1304	1414	1524	1084	1194	1304	1414	1524	CHR19A3NE474--	CHR19A3NE474--	0116
1195	1305	1415	1525	1085	1195	1305	1415	1525	CHR19A1NE684--		
1196	1306	1416	1526	1086	1196	1306	1416	1526	CHR19A1NE684--		
1197	1307	1417	1527	1087	1197	1307	1417	1527	CHR19A3NE684--		
1198	1308	1418	1528	1088	1198	1308	1418	1528	CHR19A3NE684--		
1199	1309	1419	1529	1089	1199	1309	1419	1529	CHR19A1NE105--	CHR19A1NE105--	0117
1200	1310	1420	1530	1090	1200	1310	1420	1530	CHR19A1NE105--	CHR19A1NE105--	0118
1201	1311	1421	1531	1091	1201	1311	1421	1531	CHR19A3NE105--	CHR19A3NE105--	0119
1202	1312	1422	1532	1092	1202	1312	1422	1532	CHR19A3NE105--	CHR19A3NE105--	0120
1203	1313	1423	1533	1093	1203	1313	1423	1533	CHR19A1NE155--		
1204	1314	1424	1534	1094	1204	1314	1424	1534	CHR19A1NE155--		
1205	1315	1425	1535	1095	1205	1315	1425	1535	CHR19A3NE155--		
1206	1316	1426	1536	1096	1206	1316	1426	1536	CHR19A3NE155--		
1207	1317	1427	1537	1097	1207	1317	1427	1537	CHR19A1NE225--		
1208	1318	1428	1538	1098	1208	1318	1428	1538	CHR19A1NE225--		
1209	1319	1429	1539	1099	1209	1319	1429	1539	CHR19A3NE225--		
1210	1320	1430	1540	1100	1210	1320	1430	1540	CHR19A3NE225--		
1211	1321	1431	1541	1101	1211	1321	1431	1541	CHR19A1NE335--		
1212	1322	1432	1542	1102	1212	1322	1432	1542	CHR19A1NE335--		
1213	1323	1433	1543	1103	1213	1323	1433	1543	CHR19A3NE335--		
1214	1324	1434	1544	1104	1214	1324	1434	1544	CHR19A3NE335--		

1/ Complete type designation will include additional symbols for capacitance tolerance and failure rate level.
2/ The following type designations are only listed in MIL-C-39022/2 and /2A: CHR19A1NC405--, CHR19A3NC405--, CHR19A1NC505--, CHR19A3NC505--, CHR19A1NC605--, CHR19A3NC605--, CHR19A1NC805--, CHR19A3NC805--, CHR19A1NE205--, CHR19A3NE205--, CHR19A1NE305--, and CHR19A3NE305--.

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Part or Identifying Number (PIN): M39022/02- (dash number from table I).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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